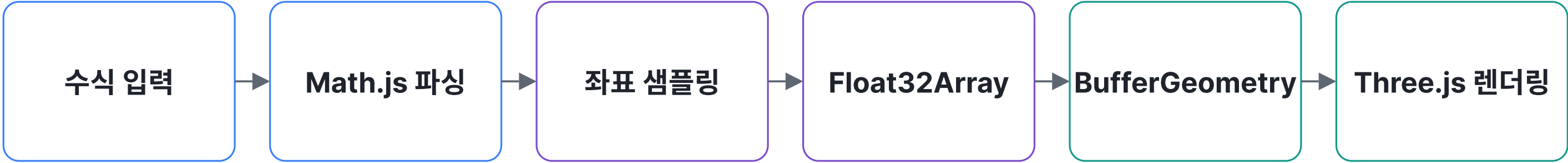


2. 수식 기반 2D·3D 그래프 렌더링



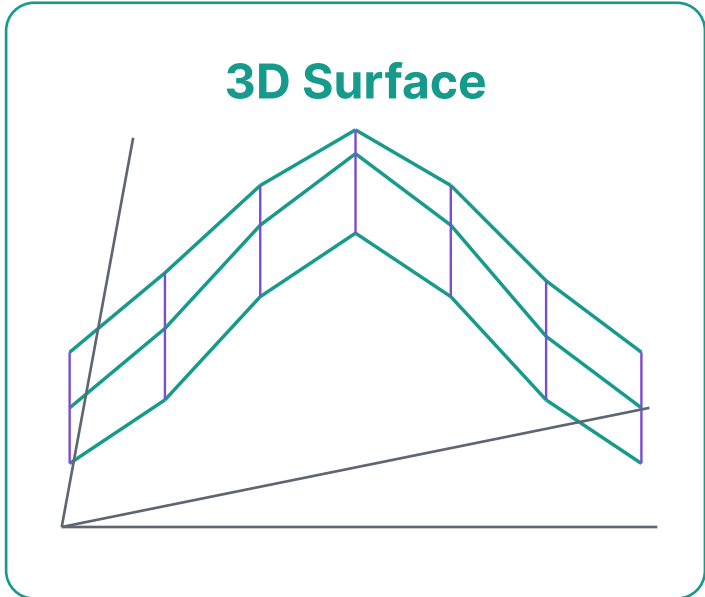
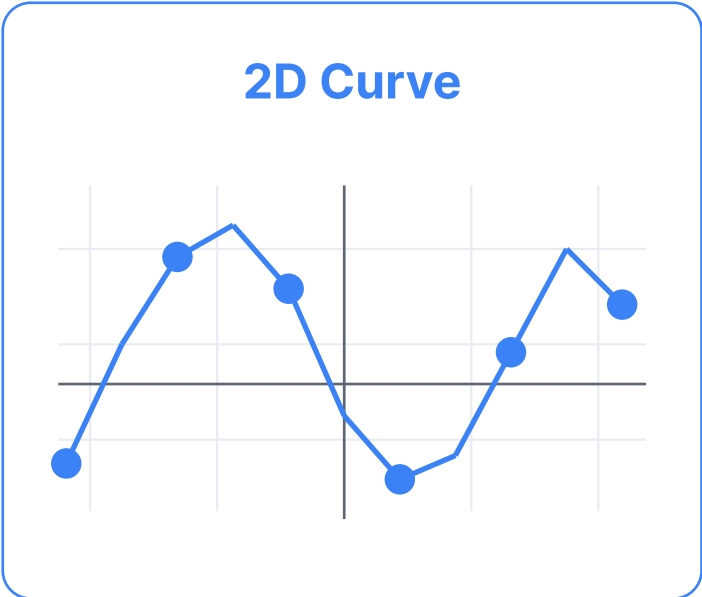
수식을 파싱하고 정의역을 샘플링해 좌표 데이터를 생성한 뒤 BufferGeometry로 변환해 2D·3D 그래프로 렌더링했습니다.

핵심 코드 구조

```
const expr = math.parse(formula);
const compiled = expr.compile();

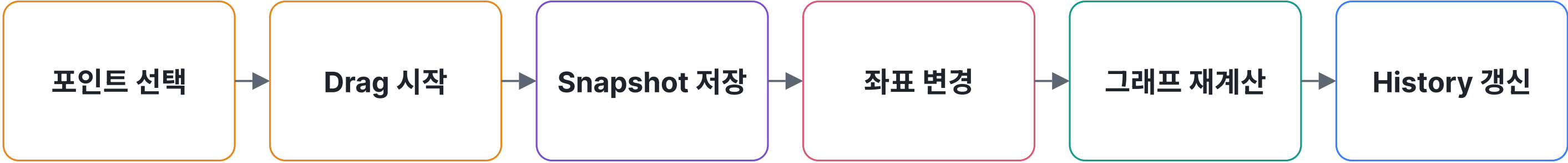
const positions =
  new Float32Array(size);

const geometry =
  new THREE.BufferGeometry();
```

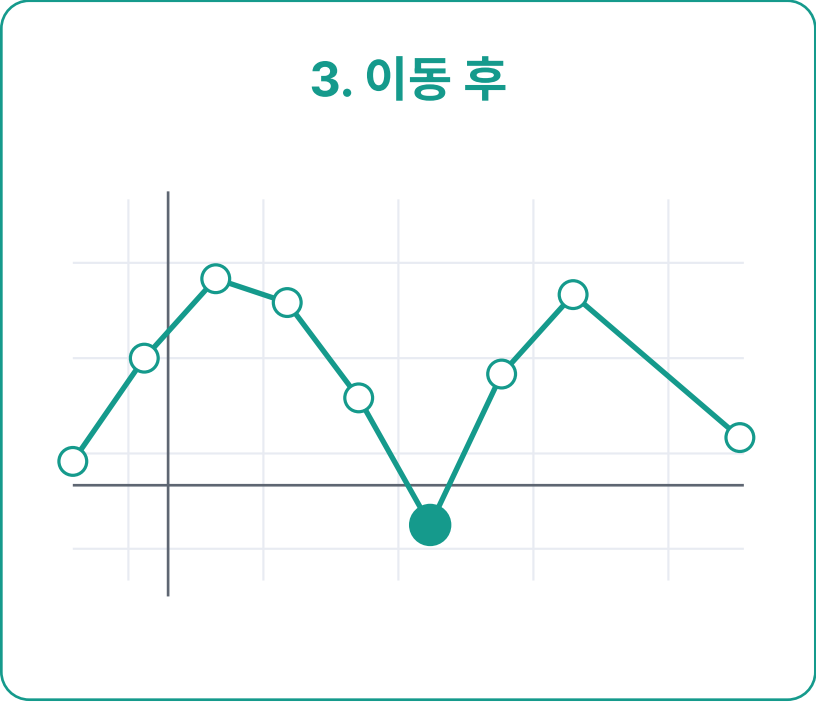
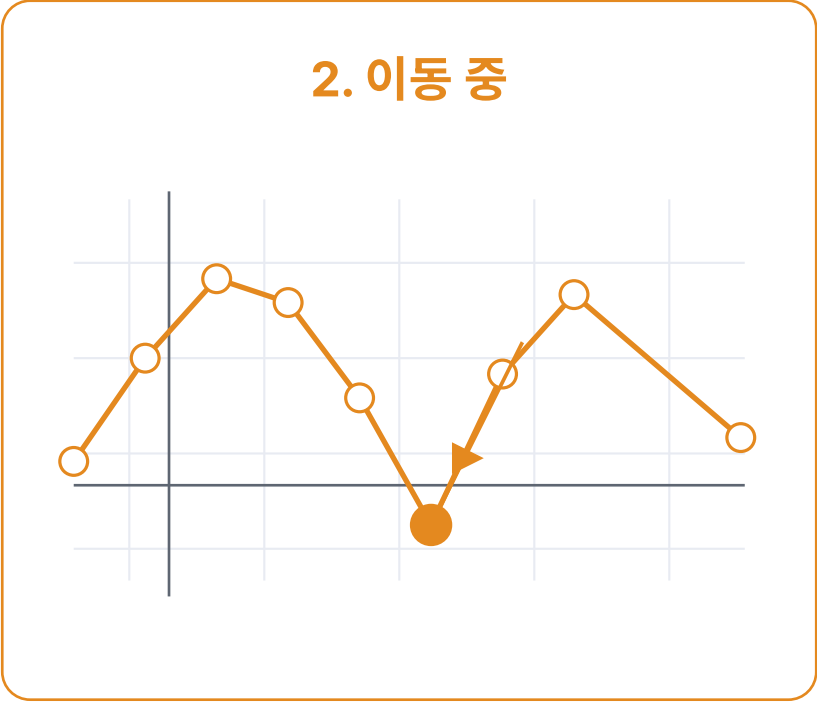
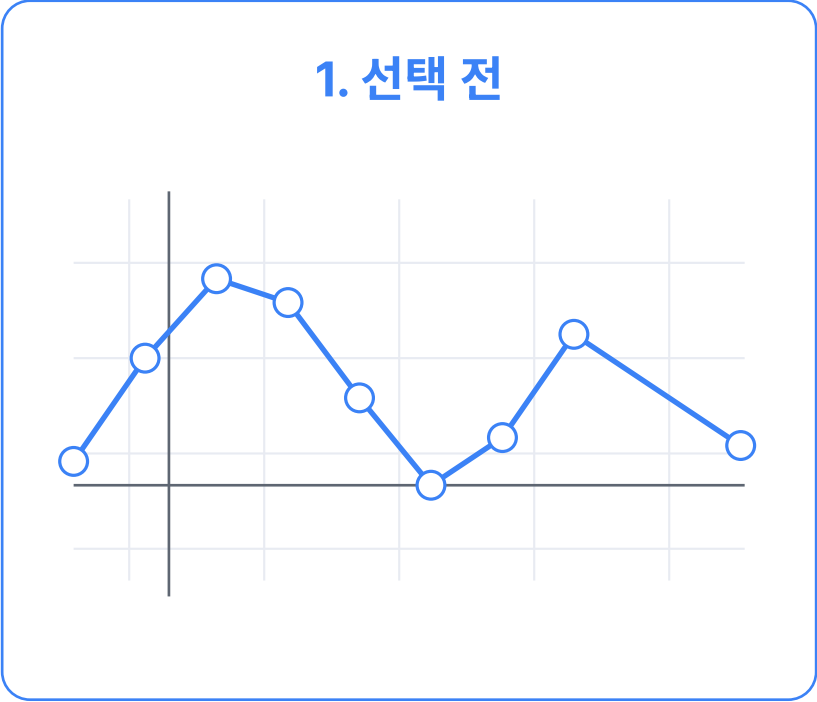


Math.js · Float32Array · BufferGeometry · React Three Fiber

3. 그래프 직접 조작 및 Undo/Redo



한 번의 Drag 동작을 하나의 Undo 단위로 관리했습니다.



TransformControls · State Snapshot · Undo / Redo

Drag 시작 시 상태 저장 → 종료 시 변경 결과를 History에 기록